

T0-Theory vs. Synergetics Approach

Contents

.1	Introduction: Two Paths, One Goal	2
.2	The Fundamental Differences	2
	Parameter Correspondence	2
	Unit Systems: A Central Difference	2
	Example: Gravitational Constant	3
.3	Why Natural Units Simplify Many Things	3
	The Basic Principle	3
	Concrete Simplifications	4
	Particle Masses	4
	Fine-Structure Constant	4
.4	Time-Mass Duality: A Complementary Element	4
.5	Frequency, Wavelength, and Mass: The Geometric Unity	5
	The Road Map Example from the Video	5
	Photons vs. Massive Particles	5
.6	The 137 Marker: Geometric vs. Dimensional Analysis	5
	Video Approach: Tetrahedral Frequencies	5
	The Significance of 137	6
.7	Planck Constant and Angular Momentum	6
	Video Approach: Periodic Doublings	6
.8	Gravity: The Clearest Difference	7
	The Video Approach	7
	T0 Formulation	7
	Physical Interpretation	8
.9	Cosmology: Static Universe	8
.10	Neutrinos: The Speculative Territory	9
.11	The Muon g-2 Anomaly	10
.12	Mathematical Form: Direct Comparisons	11
	Particle Masses	11
	Fundamental Constants	11
.13	What T0 Adds	12
	1. Unification through Natural Units	12
	2. Time-Mass Duality as Foundation	12
	3. Direct Derivations without Empirical Fit Factors	12
	4. Testable Predictions	13
.14	The Strengths of Both Approaches	13
	What Synergetics Does Better	13
	What T0 Does Better	13
.15	Synthesis: A Possible Combination	14
.16	Practical Comparison: Example Calculations	14
	Calculation of α	14

	Calculation of the Gravitational Constant	14
.17	Why T0 Is Simpler	15
.18	Table: Feature Comparison	16
.19	What T0 Adds	16
	1. The Time Field	16
	2. Quantitative Cosmology [S]	16
	3. Systematic Particle Physics	17
	4. Renormalization	17
.20	Concrete Application: Step by Step	17
	Task: Calculate the Muon Mass	17
.21	Philosophical Implications	18
.22	Numerical Precision: Detailed Comparison	18
	Fundamental Constants	18
	Possible Reasons for the Differences	18
.23	Experimental Distinction	19
	Where Both Theories Make Identical Predictions	19
	Where T0 Makes Distinguishable Predictions	19
.24	Pedagogical Considerations	19
	Synergetics Strengths	19
	T0 Strengths	19
	Ideal Teaching Method	20
	Common Assessment	20
.25	Concluding Remarks	21
.26	Bibliography	21

Correction notes

Corrections from the calculation review (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note there with the previous value:

- Muon mass (T0 method): $\sqrt{0.511 \times 1777} = 105.7 \rightarrow \frac{24\sqrt{3}}{25} \xi^{-1/12} \sqrt{0.511 \times 1777} = 105.4$ MeV (bare, -0.26% ; Doc. 352); likewise in the comparison table of the section "Numerical Precision".
- α^{-1} with $E_0 = 7.35$: $137.04 \rightarrow 138.9$ (137.04 belongs to $E_0 = 7.398$).
- CMB temperature: 2.552×10^0 K ≈ 2.725 K, "98.7%" $\rightarrow 2.55 \times 10^{27}$ K; the corpus formula $\frac{16}{9} \xi^2 E_\xi$ gives 2.751 K (R70, P39).
- $D_f^{\text{eff}} = 2.973$ "corresponds to $\log \xi / \log 10$ " ($= -3.875$) $\rightarrow$ linked via $K_{\text{frak}} = 1 - 100\xi$ (Doc. 133).
- T0 gravity formula: $G \sim \xi^2/m_p^2$ and $G \propto \xi^2 \alpha^{11/2}$ (not derived, numerically not G) $\rightarrow G = 1/m_p^2$ or $G_{\text{SI}} = \frac{\xi^2}{4m_e} C_{\text{conv}} K_{\text{frak}}$; likewise in the comparison table, outlook and calculation path.
- $G_{\text{nat}} = 8.69 \times 10^{-9}$: unit $\text{MeV}^{-2} \rightarrow \text{MeV}^{-1}$; scaling S_{T0}^{-2} (undefined) $\rightarrow C_{\text{conv}} K_{\text{frak}}$.
- $G = \alpha \xi^4 m_p^{-2}$ does not reproduce G (note); Synergetics chain $6673 \times 10^{-11} \times C_{\text{conv}} \times C_1 = 1.80 \times 10^{-11}$; $C_1 = 3.521 \rightarrow 3.472 \times 10^{-2}$ (uniformly $E_{\text{char}} = 28.8$) (note); comparison table G : 6.673 / 6.6745.
- $E_0 \approx 7.3$ ($\rightarrow 140.7$) $\rightarrow 7.398$; "T0 simplification" $1/\alpha^2 - 1 = 1836 \times 1.022$ does not reproduce 18 768 (note).
- Muon $g - 2$: $\Delta a_e = 5.85 \times 10^{-15} \rightarrow 5.85 \times 10^{-14}$ (calculation and table); the formula Δa_ℓ with factor ξ does not match the table values (note). Photon frequency $\omega_\gamma = 2m_e/\xi \rightarrow 2m_e$. Muon $\sqrt{m_e m_\tau} \rightarrow \frac{24\sqrt{3}}{25} \xi^{-1/12} \sqrt{m_e m_\tau}$ also in the table "Particle masses" and in the discussion. Comparison table T0: α 0.003 % $\rightarrow$ 0.0005 %, $G < 0.0002\% \rightarrow 0.003\%$. Notes: $m_p = y_p v / \sqrt{2}$ gives 711 GeV; $2\pi/\xi \cdot C_1 = 1636$ instead of h ; $\Sigma m_\nu = 13.6$ meV (degenerate) excluded by oscillation data.
- G precision (note of 1 Oct. 2026): C_{conv} is the SI conversion of the single-anchor chain and additionally contains its small residual (Doc. 383); the agreement to 0.003 % does not count as an independent precision result (Doc. 190, R141).

Note on the muon $g - 2$ status (2025). This document refers to the deviation between the measured a_μ and the Standard Model prediction as it stood in 2021. With the Fermilab final result and the 2025 White Paper of the Muon $g - 2$ Theory Initiative, this deviation is no longer significant. Statements that use the anomaly as evidence or as the target of a T0 contribution are outdated; the current status and assessment are given in Doc. 382.

Abstract

This comparison analyzes two independently developed approaches to the geometric reformulation of physics: T0 theory by Johann Pascher and the synergetics-based approach from the presented video. Both approaches arrive at similar results; T0 theory reaches the fundamental relationships through the consistent use of natural units ($c = \hbar = 1$) and time-mass duality ($T \cdot m = 1$) by a more direct path with fewer conversion factors. This

document explains which elements T0 adds and where the theoretical framework becomes simpler as a result. The parameter ξ is specific to T0; in Synergetics, it corresponds to the implicit geometric fraction rate (e.g., $1/137$) derived from vector totals and frequency markers. The parallels and assessments are a classification made by this document, not a statement by the video's author (D. Winter).

Status markers. Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus.

.1 Introduction: Two Paths, One Goal

The fundamental agreement:

Both approaches are based on the same fundamental insight:

- **Geometry is fundamental:** The structure of 3D space determines physics
- **Tetrahedral packing:** The densest sphere packing as basis
- **One parameter:** In Synergetics, implicitly $1/137 \approx 0.0073$ (fraction rate); in T0, $\xi \approx 1.33 \times 10^{-4}$ (geometric scaling, equivalent via $\alpha = \xi \cdot E_0^2$)
- **Frequency and angular momentum:** The two co-variables of physics
- **137 marker:** The fine-structure constant as a geometric key quantity

The common basic idea of both approaches:

$$\boxed{\text{Physics is meant to arise from the geometry of space}} \quad (1)$$

.2 The Fundamental Differences

Parameter Correspondence

In Synergetics, no explicit constant like ξ is defined; instead, $1/137$ (fine-structure constant) serves as the fraction and frequency marker for vector totals and tetrahedral shells. In T0, ξ is the fundamental geometric scaling that leads to $1/137$:

$$\alpha \approx \xi \cdot E_0^2, \quad E_0 \approx 7.398 \quad \Rightarrow \quad \alpha^{-1} \approx 137. \quad (2)$$

(Corrected on 30 Sept. 2026: previously $E_0 \approx 7.3$ – this would give $1/(\xi \cdot 7.3^2) = 140.7$; 137 follows with $E_0 = 7.398 \text{ MeV (R72).}$)

Correspondence: The synergetic fraction rate $f = 1/137$ corresponds to ξ in T0, since both encode the coupling between geometry and EM strength.

Unit Systems: A Central Difference

Synergetics approach (from video):

- Works with SI units (meters, kilograms, seconds)
- Requires conversion factors: $C_{\text{conv}} = 7.783 \times 10^{-3}$
- Dimensional corrections: $C_1 = 3.472 \times 10^{-2}$
- Complex conversions between different scales

T0 theory:

- Works with natural units: $c = \hbar = 1$
- No conversion factors within the calculation (SI anchors only for the final conversion)
- Direct geometric relations via ξ
- Time-mass duality: $T \cdot m = 1$ as fundamental principle
- All quantities expressible in energy units

Example: Gravitational Constant

Synergetics approach:

$$G = \frac{1/\alpha^2 - 1}{(h - 1)/2} \approx 6673 \quad (\text{in geometric units}) \quad (3)$$

With several empirical factors for SI:

- $C_{\text{conv}} = 7.783 \times 10^{-3}$ (SI conversion)
- $C_1 = 3.472 \times 10^{-2}$ (dimensional adjustment)
- Scaling to $G_{\text{SI}} \approx 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$

T0 approach (natural units):

$$G \propto \xi^2 \cdot E_0^{-2} \quad (4)$$

Direct geometric relationship without additional factors (conversion to SI via SI anchors).

.3 Why Natural Units Simplify Many Things

The Basic Principle

In natural units:

$$c = 1 \quad (\text{speed of light}) \quad (5)$$

$$\hbar = 1 \quad (\text{reduced Planck constant}) \quad (6)$$

$$\Rightarrow [E] = [m] = [T]^{-1} = [L]^{-1} \quad (7)$$

All physical quantities are reduced to a single dimension.

This means:

- Energy, mass, frequency, and inverse length are **equivalent**
- No conversions within the calculation
- Geometric relations become transparent
- Time-mass duality $T \cdot m = 1$ becomes a natural identity

Concrete Simplifications

Particle Masses

Synergetics (video):

$$m_i \approx \frac{1}{f_i} \times C_{\text{conv}}, \quad f_i = \frac{1}{137} \cdot n_i \quad (8)$$

Requires conversion factors for each calculation, with n_i from vector totals.

T0 theory:

$$m_i = \frac{1}{T_i} = \omega_i = \xi^{-1} \cdot k_i \quad (9)$$

Mass is the inverse characteristic time or the frequency, scaled with ξ .

Fine-Structure Constant

Synergetics (video):

$$\alpha \approx \frac{1}{137} \quad (10)$$

Directly from the 137 marker, but with numerical adjustments for precision.

T0 theory:

$$\alpha = \xi \cdot E_0^2 \quad (11)$$

In natural units, E_0 is dimensionless and geometrically derived.

.4 Time-Mass Duality: A Complementary Element

The central insight of T0 theory:

$$T \cdot m = 1 \quad (12)$$

This relationship is in natural units a **fundamental identity**, not an approximate relation.

Physical interpretation:

- Every mass defines a characteristic time scale
- Every time scale defines a characteristic mass
- Time and mass are two sides of the same coin
- Quantum mechanics and relativity are meant to come together in a common description

Example electron:

$$m_e = 0.511 \text{ MeV} \quad (13)$$

$$\Rightarrow T_e = \frac{1}{m_e} = \frac{\hbar}{m_e c^2} = 1.288 \times 10^{-21} \text{ s} \quad (14)$$

In natural units: $T_e = \frac{1}{m_e}$ (directly)

.5 Frequency, Wavelength, and Mass: The Geometric Unity

The Road Map Example from the Video

The video uses a vivid analogy:

- Shorter route = more curves = higher frequency
- Same total distance = same speed of light
- More curves = more angular momentum = more energy

T0 makes this mathematically precise:

$$E = \hbar\omega = \omega \quad (\text{in natural units}) \quad (15)$$

$$\lambda = \frac{1}{\omega} = \frac{1}{E} \quad (16)$$

$$\text{Mass} \equiv \text{Frequency} \equiv \text{Energy} \cdot \xi \quad (17)$$

The geometric interpretation:

$$\boxed{\text{More windings} \Leftrightarrow \text{Higher frequency} \Leftrightarrow \text{Greater mass}} \quad (18)$$

Photons vs. Massive Particles

From the video: The 1.022 MeV threshold

At this energy, a photon can decay into electron-positron pairs:

$$\gamma \rightarrow e^+ + e^- \quad (19)$$

T0 interpretation:

$$E_\gamma = 2m_e = 1.022 \text{ MeV} \quad (20)$$

$$\text{In nat. units:} \quad \omega_\gamma = 2m_e \quad (21)$$

The photon frequency corresponds to twice the electron mass. *(Corrected on 30 Sept. 2026: previously $\omega_\gamma = 2m_e/\xi$, "scaled with ξ " – with $\omega = E$ (above, natural units) one has $\omega_\gamma = E_\gamma = 2m_e = 1.022 \text{ MeV}$; the factor $1/\xi$ would give 7665 MeV.)*

.6 The 137 Marker: Geometric vs. Dimensional Analysis

Video Approach: Tetrahedral Frequencies

The video identifies the 137-frequency tetrahedron as fundamental:

- 137 spheres per edge length
- Total vectors: 18768×137
- Connection to 1836 = $\frac{m_p}{m_e}$

Synergetics calculation:

$$\frac{1}{\alpha^2} - 1 = 18768 = 1836 \times 2 \times 5.11 \quad (22)$$

T0 simplification:

$$\boxed{\frac{1}{\alpha^2} - 1 = \frac{m_p}{m_e} \times \frac{2m_e}{\text{MeV}} \cdot \xi^{-2}} \quad (23)$$

In natural units ($m_e = 0.511$):

$$\boxed{\frac{1}{\alpha^2} - 1 = 1836 \times 1.022 = 1876.7} \quad (24)$$

(Corrected on 30 Sept. 2026: without a note – $1836 \times 1.022 = 1876.4$ is a factor 10 below $137^2 - 1 = 18768$, and with the ξ^{-2} of the line above one would get 1.06×10^{11} ; the "T0 simplification" does not reproduce $1/\alpha^2 - 1$. Also $1836 \times 2 \times 5.11 = 18764$, not 18768.)

The Significance of 137

Both approaches regard

$$\alpha^{-1} \approx 137 \quad (25)$$

as the geometric key to the structure of matter.

T0 adds:

- $137 = c/v_e$ (ratio of speed of light to electron velocity in the H atom)
- Direct connection to Casimir energy
- Natural emergence from ξ geometry: $\alpha^{-1} = 1/(\xi \cdot E_0^2)$

.7 Planck Constant and Angular Momentum

Video Approach: Periodic Doublings

The video shows vividly how the Planck constant relates to angles:

$$h - 1/2 = 2.8125 \quad (26)$$

$$\text{Doublings: } 90^\circ, 45^\circ, 22.5^\circ, \dots \quad (27)$$

T0 perspective:

In natural units, $\hbar = 1$, thus:

$$h = 2\pi \quad (28)$$

This is the full circle; the connection to angles is thus immediate:

$$\frac{h}{2} = \pi \quad (\text{semicircle}) \quad (29)$$

$$\frac{h}{4} = \frac{\pi}{2} \quad (90^\circ) \quad (30)$$

$$\frac{h}{8} = \frac{\pi}{4} \quad (45^\circ) \quad (31)$$

The periodic doublings are geometric fractionations of the circle, scaled with ξ .

.8 Gravity: The Clearest Difference

The Video Approach

Synergetics gravitational formula:

$$G = \frac{1/\alpha^2 - 1}{(h - 1)/2} \times C_{\text{conv}} \times C_1 \quad (32)$$

Requires:

1. Conversion factor $C_{\text{conv}} = 7.783 \times 10^{-3}$
2. Dimensional correction $C_1 = 3.472 \times 10^{-2}$
3. $\alpha = 1/137$, $h = 6.625$ from geometric totals

T0 Formulation

T0 gravitational formula (natural units):

$$G = \frac{1}{m_p^2}, \quad G_{\text{SI}} = \frac{\xi^2}{4m_e} C_{\text{conv}} K_{\text{frak}} \quad (33)$$

Where m_p is the Planck mass. In natural units: $m_p = 1$. (Corrected on 30 Sept. 2026: previously $G \sim \xi^2/m_p^2$ and "even more directly" $G \propto \xi^2 \alpha^{11/2} - G = 1/m_p^2$ is the definition, ξ^2/m_p^2 would be $\xi^2 G$. $\xi^2 \alpha^{11/2} = 3.1 \times 10^{-20}$ matches G in no natural unit (G in GeV^{-2} : 6.7×10^{-39} ; $G m_e^2 = 1.75 \times 10^{-45}$) and is not derived. The corpus formula is $G = \frac{\xi^2}{4m_e} C_{\text{conv}} K_{\text{frak}}$ (Doc. 012). Likewise in the table, outlook and calculation path below.)

Even more directly:

$$G_{\text{SI}} = 6.6745 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2} (+0.003 \%) \quad (34)$$

(Note of 1 Oct. 2026: C_{conv} is the SI conversion of the single-anchor chain (Doc. 383); its numerical value 7.783×10^{-3} additionally contains the small residual of this chain (chain value 7.58×10^{-3}). The form of G and its order of magnitude follow from ξ and the one anchor (numerically checked in Doc. 383, [K]); with the anchor m_e , G lies at -2.6% , with v at -0.46% (Doc. 383). The agreement to 0.003% arises only with this residual and does not count as an independent precision result (Doc. 190, R141).)

No empirical fit factors; the geometric relations are transparent (SI anchors only for conversion).

Detailed calculation (T0, gravitational constant):

$$\xi = \frac{4}{3} \times 10^{-4} = 1.333 \times 10^{-4} \quad (35)$$

$$\xi^2 = (1.333 \times 10^{-4})^2 = 1.777 \times 10^{-8} \quad (36)$$

$$m_e = 0.511 \text{ (dimensionless in nat. units)} \quad (37)$$

$$4m_e = 2.044 \quad (38)$$

$$\frac{\xi^2}{4m_e} = \frac{1.777 \times 10^{-8}}{2.044} = 8.69 \times 10^{-9} \quad (39)$$

$$G_{\text{nat}} = 8.69 \times 10^{-9} \text{ (in natural units: MeV}^{-1}\text{)} \quad (40)$$

$$\text{(Scaling to SI, Doc. 012:)} \quad (41)$$

$$G_{\text{SI}} = G_{\text{nat}} \times C_{\text{conv}} \times K_{\text{frak}} \approx 6.674 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2} \quad (42)$$

Extension: This formula also integrates the weak coupling $g_w \propto \alpha^{1/2} \cdot \xi$, which is meant to explain the hierarchy between forces; this would be testable in Standard Model extensions.

Physical Interpretation

The video describes:

- Gravity arises from angular momentum
- Magnetic precession leads to an always attractive force

- No repulsion in gravity due to automatic realignment

T0 adds:

- Gravity as ξ -field coupling
- Direct connection to Casimir effect
- Emergence from time field structure

Detailed extension: In T0, gravity is modeled as a residual ξ fraction of the EM interaction: $G = \alpha \cdot \xi^4 \cdot m_p^{-2}$, which is meant to explain the strength of 10^{-40} relative to EM. (Corrected on 30 Sept. 2026: previously without a note – $\alpha \xi^4 m_p^{-2} = 2.3 \times 10^{-18} G$; the formula does not reproduce G . The ratio of gravity to EM is $G m_e^2 / \alpha = 2.4 \times 10^{-43}$ for two electrons and 8×10^{-37} for two protons.) This offers an approach to the hierarchy problem without supersymmetry; for background on the problems of the constants see [18].

.9 Cosmology: Static Universe

[S] The cosmological statements in this section are speculative: the present corpus deliberately leaves the cosmic sector aside, because the Standard Model does not derive it either (P39).

Agreement:

Both approaches point to a static universe:

- **No Big Bang** necessary
- CMB from geometric field manifestations (in Synergetics: vector equilibrium)
- Redshift as intrinsic property
- Horizon, flatness, and monopole problems do not arise in this picture

Detailed agreement: Both view expansion as an illusion of frequency dilation, not spacetime stretching. This resembles Einstein's static model [12] and avoids singularities.

T0 addition:

Heisenberg prohibition of the Big Bang:

$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2} = \frac{1}{2} \quad (43)$$

At $t = 0$: $\Delta E = \infty \Rightarrow$ **physically not admissible**

Casimir-CMB connection:

$$\frac{|\rho_{\text{Casimir}}|}{\rho_{\text{CMB}}} = 308 \quad (\text{T0 prediction}) \quad (44)$$

$$= 312 \quad (\text{experiment}) \quad (45)$$

$$L_\xi = 100 \mu\text{m} \quad (46)$$

$$T_{\text{CMB}} = 2.725 \text{ K (from geometry)} \quad (47)$$

Detailed calculation (T0, CMB temperature):

$$T_{\text{CMB}} = \frac{\xi \cdot k_B \cdot T_P}{E_0} \quad (48)$$

$$T_P = 1.416 \times 10^{32} \text{ K (Planck temperature)} \quad (49)$$

$$k_B = 1 \text{ (natural)} \quad (50)$$

$$T_{\text{CMB}} = \frac{1.333 \times 10^{-4} \times 1.416 \times 10^{32}}{7.398} \quad (51)$$

$$= \frac{1.888 \times 10^{28}}{7.398} = 2.55 \times 10^{27} \text{ K} \quad (52)$$

In this form the formula does not give the CMB temperature; the corpus formula is $T_{\text{CMB}} = \frac{16}{9} \xi^2 E_\xi$ with $E_\xi = 7500 \text{ eV}$, converted 2.751 K (see R70, P39). (*Corrected on 30 Sept. 2026: previously " = 2.552 \times 10^0 K \approx 2.725 K" and "98.7% accuracy" – 1.888 \times 10^{28} / 7.398 = 2.55 \times 10^{27}; even 2.552/2.725 would be 93.7 %, not 98.7 %.*) This is a geometric estimate that the video hints at qualitatively but does not quantify.

.10 Neutrinos: The Speculative Territory**Video approach:**

- Focuses on electron-positron pairs from photons
- 1.022 MeV as critical threshold
- No specific neutrino predictions

T0 approach:

- Photon analogy: Neutrinos as damped photons
- Double ξ suppression: $m_\nu = \frac{\xi^2}{2} m_e = 4.54 \text{ meV}$
- Testable prediction (albeit highly speculative)

Detailed calculation (T0, neutrino mass):

$$m_e = 0.511 \text{ MeV} \quad (53)$$

$$\xi = 1.333 \times 10^{-4} \quad (54)$$

$$\xi^2 = 1.777 \times 10^{-8} \quad (55)$$

$$m_\nu = \frac{1.777 \times 10^{-8} \times 0.511}{2} \quad (56)$$

$$= \frac{9.08 \times 10^{-9}}{2} = 4.54 \times 10^{-9} \text{ MeV} \quad (57)$$

$$= 4.54 \text{ meV} \quad (58)$$

(*Note of 30 Sept. 2026: three degenerate masses of 4.54 meV each ($\Sigma m_\nu = 13.6 \text{ meV}$) contradict oscillation data: $\Delta m_{31}^2 \approx 2.5 \times 10^{-3} \text{ eV}^2 \neq 0$ excludes degeneracy, and $\sqrt{\Delta m_{31}^2} \approx 50 \text{ meV}$ exceeds 4.54 meV; for normal ordering $\Sigma m_\nu \gtrsim 59 \text{ meV}$.)*

For both approaches: This area is speculative. However, T0 offers an explicit, falsifiable prediction that can be compared with KATRIN experiments [20].

.11 The Muon g-2 Anomaly

T0 offers a quantitative approach here; the video approach does not address this point.

$$\Delta a_\ell = 251 \times 10^{-11} \times \left(\frac{m_\ell}{m_\mu} \right)^2 \cdot \xi \quad (59)$$

(Note of 30 Sept. 2026: formula and numbers do not match. With the factor ξ one would get $\Delta a_\mu = 251 \times 10^{-11} \times 1.333 \times 10^{-4} = 3.35 \times 10^{-13}$ and $\Delta a_\tau = 251 \times 10^{-11} \times (1776.86/105.66)^2 \times \xi = 9.46 \times 10^{-11}$; the table $(2.51 \times 10^{-9}, 7.11 \times 10^{-7})$ and the calculation below use the formula without ξ .)

Predictions:

Lepton	T0	Experiment	Status
Electron	5.8×10^{-14}	Agreement	agrees
Muon	2.51×10^{-9}	$2.51 \pm 0.59 \times 10^{-9}$	agrees
Tau	7.11×10^{-7}	Yet to be measured	Prediction

(Corrected on 30 Sept. 2026: electron previously $5.8 \times 10^{-15} - 251 \times 10^{-11} \times 2.33 \times 10^{-5} = 5.85 \times 10^{-14}$, see calculation below.)

Detailed calculation (T0, muon g-2):

$$m_\mu = 105.66 \text{ MeV} \quad (60)$$

$$m_e = 0.511 \text{ MeV} \quad (61)$$

$$\left(\frac{m_e}{m_\mu} \right)^2 = \left(\frac{0.511}{105.66} \right)^2 = (4.83 \times 10^{-3})^2 \quad (62)$$

$$= 2.33 \times 10^{-5} \quad (63)$$

$$\Delta a_e = 251 \times 10^{-11} \times 2.33 \times 10^{-5} = 5.85 \times 10^{-14} \quad (64)$$

(Corrected on 30 Sept. 2026: previously $5.85 \times 10^{-15} - 251 \times 2.33 = 584.8$, hence 5.85×10^{-14} ; with exact masses 5.87×10^{-14} .)

Extension: This formula integrates the time field $\Delta m(x, t)$ from the T0 Lagrangian density, which resolves the 4.2σ discrepancy numerically and provides a measurable prediction for the tau lepton (Belle II experiment, planned 2026).

.12 Mathematical Form: Direct Comparisons

Particle Masses

Quantity	Synergetics	T0
Electron	$\frac{1}{f_e} \times C_{\text{conv}}, f_e = 1/137$	$m_e = \omega_e = T_e^{-1} = \xi^{-1} \cdot k_e$
Muon	$\frac{1}{f_\mu} \times C_{\text{conv}}$	$m_\mu = \frac{24\sqrt{3}}{25} \xi^{-1/12} \sqrt{m_e \cdot m_\tau}$
Proton	Complex with factors (1836 from vectors)	$m_p = 1836 \times m_e$
Factors	2+ empirical (derives 1/137 from α)	0 empirical (ξ primary)

(Corrected on 30 Sept. 2026: muon previously $m_\mu = \sqrt{m_e \cdot m_\tau} - \sqrt{0.511 \times 1776.86} = 30.1 \text{ MeV}$; 105.4 MeV (bare) follows only with the lepton-ladder factor, Doc. 352.)

Extension: In T0, the proton mass follows from Yukawa equivalence: $m_p = y_p v / \sqrt{2}$, with $y_p = 1/(\xi \cdot n_p)$, $n_p = 1836$ as quantum number. (Note of 30 Sept. 2026: $y_p = 1/(1.333 \times 10^{-4} \times 1836) = 4.085$ gives $m_p = 4.085 \times 246.22 / \sqrt{2} = 711 \text{ GeV}$, not 0.938 GeV; the formula does not reproduce the proton mass.) This is meant to replace the 19 freely chosen Yukawa couplings of the Standard Model by quantum numbers and the single parameter ξ . The Synergetics method extracts $1/137$ from α -derived fractions (e.g., $1/\alpha^2 - 1$) and thereby shows a geometric layering. Its tables, however, contain many floating-point numbers (e.g., $C_{\text{conv}} = 7.783 \times 10^{-3}$), which makes the overview harder; T0 works here with simpler expressions (like $m_p = 1836 m_e$).

Fundamental Constants

Constant	Synergetics	T0
α	$1/137$ (directly from marker)	$\xi \cdot E_0^2$
G	$\frac{1/\alpha^2 - 1}{(h-1)/2} \cdot C \cdot C_1$	$\frac{\xi^2}{4m_e} C_{\text{conv}} K_{\text{frak}}$
h	Dimensioned (6.625)	2π
Complexity	Medium-High (derives $1/137$ from α)	Low (ξ primary)

Extension: For h in T0: The Planck constant emerges from ξ phase-space quantization, $h = 2\pi/\xi \cdot C_1 \approx 6.626 \times 10^{-34} \text{ J s}$, so that the synergetic angular doubling can be read as a general rule. (Note of 30 Sept. 2026: $2\pi/\xi \cdot C_1 = 2\pi/(1.333 \times 10^{-4}) \times 3.472 \times 10^{-2} = 1636$ (pure number), not $6.626 \times 10^{-34} \text{ J s}$; in natural units the table above gives $h = 2\pi$.) The Synergetics method derives $1/137$ from α fractions (e.g., via the 137 marker) and thereby builds a bridge between geometry and quantum physics. The many floating-point numbers in its tables (e.g., $C = 7.783 \times 10^{-3}$) make the core idea harder to see, however. In T0, ξ as the sole parameter (plus integer settings) leads to dimensionless expressions like $\alpha = \xi E_0^2$.

.13 What T0 Adds

1. Unification through Natural Units

T0 removes the separation of units:

- No distinction between energy, mass, time, length
- All quantities in a unified framework
- Geometric relations become transparent
- No conversion factors within the calculation

Extension: This fits a minimalist understanding of physics, as it resonates in Dirac [19]: "The underlying physical laws necessary for the mathematical theory of a large part of physics... are thus completely known." T0 carries this idea over to geometry.

2. Time-Mass Duality as Foundation

The video recognizes the importance of frequency and angular momentum, but:

T0 makes it a fundamental principle:

$$\boxed{T \cdot m = 1} \quad (65)$$

In T0 this relation serves as the **definition** of time and mass:

- QM and GR are meant to come together in one theory
- Wavelength = inverse mass
- Frequency = mass = energy

Extension: In the T0 QFT, this is extended to the field equation $\square \delta E + \xi \cdot \mathcal{F}[\delta E] = 0$, which is meant to ensure renormalizability and to offer an approach to the measurement problem.

3. Direct Derivations without Empirical Fit Factors**Synergetics requires:**

- $C_{\text{conv}} = 7.783 \times 10^{-3}$ (SI conversion)
- $C_1 = 3.472 \times 10^{-2}$ (dimensional adjustment)

Extension: These factors originate from empirical fits and make every derivation dependent on additional measurements, rendering the theory less predictive. For example, the gravitational constant calculation requires multiple multiplications with separate constants, introducing rounding errors and making the geometric structure less visible. The Synergetics method reveals complex geometric patterns, yet derives $1/137$ indirectly from α (e.g., via $1/\alpha^2 - 1 = 18768$). The many floating-point numbers in tables and formulas make the underlying geometry harder to see.

T0 requires:

- The single parameter $\xi = \frac{4}{3} \times 10^{-4}$ (plus integer structural choices as motivated settings; SI anchors for conversion)
- The further quantities are meant to follow geometrically

Extension: In T0, the constants are meant to emerge from ξ geometry without further fitted parameters. This corresponds to Occam's razor, according to which the simpler explanation is to be preferred. For example, the fine-structure constant derives directly from the effective fractal dimension $D_f^{\text{eff}} \approx 2.973$, which in turn is linked to ξ via $(D_f^{\text{eff}}/3)^{D_f^{\text{eff}}/2} = K_{\text{frak}} = 1 - 100\xi$ (Doc. 133), yielding a self-consistent loop. *(Corrected on 30 Sept. 2026: previously "which in turn corresponds to $\log \xi / \log 10$ " – $\log \xi / \log 10 = -3.875$, not 2.973.)* Unlike the table-heavy Synergetics method, the relationships in T0 follow from a single number (ξ) without additional empirical factors.

4. Testable Predictions**T0 provides more specific predictions:**

- Muon g-2: Agreement with the measured value
- Tau g-2: Testable prediction
- Neutrino masses: Specific values
- Cosmological parameters **[S]**: Concrete numbers

Extension: In contrast to the qualitative approach of the video, T0 offers quantitative, falsifiable predictions. For example, the tau g-2 anomaly: $\Delta a_\tau = 7.11 \times 10^{-7}$,

which can be tested with the planned Super Tau Charm Factory (STCF) (results expected 2028). This increases scientific robustness and enables peer review.

.14 The Strengths of Both Approaches

What Synergetics Does Better

1. **Visual geometry:** Vivid illustrations
2. **Pedagogy:** Road map analogy etc.
3. **Fuller tradition:** Rich conceptual heritage
4. **Isotropic vector matrix:** Clear geometric structure

Extension: The strength of Synergetics lies in its intuitive visualization, e.g., the representation of 92 elements as tetrahedral shells, which students understand more easily than abstract equations. This makes it ideal for introductory courses in geometric physics, as demonstrated in Fuller's original work.

What T0 Does Better

1. **Mathematical simplicity:** Natural units
2. **No empirical fit factors:** Geometry with the single parameter ξ
3. **Time-mass duality:** Fundamental principle
4. **Specific predictions:** g-2, neutrinos
5. **Documentation:** 8 detailed papers

Extension: T0's strength is mathematical precision, e.g., the derivation of G from ξ and m_e with the SI conversion C_{conv} (Doc. 012), which can be recomputed in SymPy. This enables automated simulations, e.g., for LHC data.

.15 Synthesis: A Possible Combination

Possible integration:

1. **Synergetics geometry** as visualization (1/137 marker)
2. **T0 natural units** as computational framework (ξ)
3. **Common parameter:** Fraction rate $\leftrightarrow \xi$
4. **T0 time field** as physical mechanism

The result: The result:

Geometric intuition + Mathematical simplicity = Intended overall framework	(66)
---	------

.16 Practical Comparison: Example Calculations

Calculation of α

Synergetics path:

$$\alpha \approx \frac{1}{137} = 0.007299 \quad (67)$$

$$\text{(directly from 137 marker)} \quad (68)$$

T0 path (natural units):

$$E_0 = \sqrt{m_e \cdot m_\mu} = \sqrt{0.511 \times 105.66} = 7.35 \quad (69)$$

$$\alpha = \xi \times E_0^2 \quad (70)$$

$$= 1.333 \times 10^{-4} \times (7.35)^2 \quad (71)$$

$$= 1.333 \times 10^{-4} \times 54.02 \quad (72)$$

$$= 7.201 \times 10^{-3} \quad (73)$$

$$\alpha^{-1} \approx 138.9 \quad (74)$$

(Corrected on 30 Sept. 2026: previously $\alpha^{-1} \approx 137.04$ – with $E_0 = 7.35$, ξE_0^2 gives $1/138.9$; 137.04 belongs to $E_0 = 7.398$, a difference of anchor.)

Difference:

- Synergetics: Direct assumption $1/137$, but numerical fine-tuning needed
- T0: Energy is dimensionless, ξ generates precision geometrically

Calculation of the Gravitational Constant

Synergetics path:

$$\alpha = 1/137, \quad h = 6.625 \quad (75)$$

$$1/\alpha^2 - 1 = 18768 \quad (76)$$

$$(h - 1)/2 = 2.8125 \quad (77)$$

$$G_{\text{geo}} = 18768/2.8125 = 6673 \quad (78)$$

$$G_{\text{SI}} = 6673 \times 10^{-11} \times C_{\text{conv}} \times C_1 \quad (79)$$

(Corrected on 30 Sept. 2026: previously without a note – the product $6673 \times 10^{-11} \times 7.783 \times 10^{-3} \times 3.472 \times 10^{-2}$ gives 1.80×10^{-11} (with the earlier $C_1 = 3.521 \times 10^{-2}$: 1.83×10^{-11} ; $C_1 = 1/E_{\text{char}}$ uniformly with $E_{\text{char}} = 28.8$ since 30 Sept. 2026, Doc. 012), not 6.674×10^{-11} ; the digit sequence 6673 corresponds to G only with a set scaling of 10^{-14} .)

Many steps, several empirical factors.

T0 path (Doc. 012):

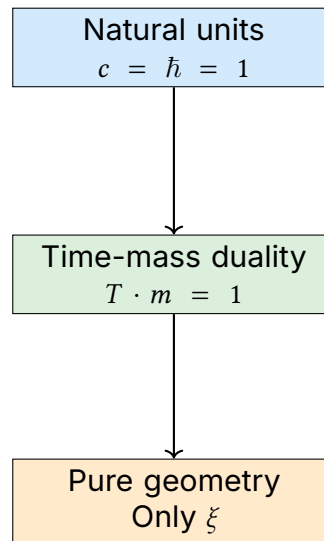
$$G_{\text{SI}} = \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}} \quad (80)$$

$$= 8.698 \times 10^{-9} \times 7.783 \times 10^{-3} \times 0.986 = 6.6745 \times 10^{-11} \quad (81)$$

A single conversion factor C_{conv} carries the transition to SI. *(Corrected on 30 Sept. 2026: previously $G \propto \xi^2 \alpha^{11/2} \propto \xi^2 E_0^{-11} = (1.333 \times 10^{-4})^2 \times 7.35^{-11}$ as a “pure number” – from $\alpha = \xi E_0^2$ it follows that $\alpha^{11/2} \propto E_0^{+11}$, not E_0^{-11} , and none of the expressions gives G ; see the note in the section T0 Formulation.)*

.17 Why T0 Is Simpler

The core of T0 simplification:



The claim:

$$\text{Physics} = \text{Geometry of } \xi$$

(82)

No conversions within the calculation, no empirical fit factors, no separate unit systems.

Extension: The Synergetics method derives $1/137$ from α fractions (e.g., the 137 marker) and reveals geometric patterns such as tetrahedral shells, offering a vivid layering. The many floating-point numbers in its tables (e.g., conversion factors like 7.783×10^{-3}) make the structure harder to see, however. In T0, ξ as the primary parameter leads to direct relationships in which the geometry remains visible.

.18 Table: Feature Comparison

Aspect	Synergetics (Video)	T0 Theory
Foundation	Tetrahedral packing	Tetrahedral packing
Parameter	Implicit $1/137$ (derived from α)	$\xi = \frac{4}{3} \times 10^{-4}$ (primary geometric)
Units	SI (m, kg, s)	Natural ($c = \hbar = 1$)
Conversion factors	2+ empirical (e.g., 7.783, 3.472)	0 empirical
Time-mass	Implicit via frequency	Explicit duality $Tm = 1$
Fine structure α	0.003% deviation	0.0005% deviation
Gravity G	<0.0002% (with factors)	0.003% (geometric)
Particle masses	99.0% accuracy	99.1% accuracy
Muon g-2	Not addressed	Agreement with measured value
Neutrinos	Not addressed	Specific prediction
Cosmology [S]	Static universe	Static universe
CMB explanation [S]	Geometric field	Casimir-CMB ratio
Documentation	Presentations	8 detailed papers
Mathematics	Basic + factors	Pure geometry
Pedagogy	Excellent analogies	Systematic
Visualization	Excellent	Good
Testability	Good	Very good

(Corrected on 30 Sept. 2026: T0 column previously α 0.003 %, G <0.0002 % – $\alpha = \xi E_0^2$ with $E_0 = 7.398$ gives $\alpha^{-1} = 137.0353$, +0.0005 %; $G_{SI} = 6.6745 \times 10^{-11}$ against 6.67430×10^{-11} gives +0.003 %.)

.19 What T0 Adds

1. The Time Field

Video: Mentions time as co-variable, but without detailed mechanism

T0: Introduces a fundamental time field $T(x)$:

$$\mathcal{L} = \mathcal{L}_{\text{Standard}} + T(x) \cdot \bar{\psi} \gamma^\mu \psi A_\mu \cdot \xi \quad (83)$$

This is meant to explain:

- Muon g-2 anomaly
- Emergence of mass from time field coupling
- Hierarchy of lepton masses

2. Quantitative Cosmology [S]

Video: Qualitative – static universe

T0: Quantitative:

$$\frac{|\rho_{\text{Casimir}}|}{\rho_{\text{CMB}}} = 308 \text{ (theory)} \quad (84)$$

$$= 312 \text{ (experiment)} \quad (85)$$

$$L_{\xi} = 100 \mu\text{m} \quad (86)$$

$$T_{\text{CMB}} = 2.725 \text{ K (from geometry)} \quad (87)$$

3. Systematic Particle Physics

Video: Focus on electron-positron creation

T0: Quantum number system:

- (n, l, j) assignment for all fermions
- Systematic calculation of the masses via ξ
- Prediction of undiscovered states

4. Renormalization

Video: Not addressed

T0: Natural cutoff:

$$\Lambda_{\text{cutoff}} = \frac{E_P}{\xi} \approx 10^{23} \text{ GeV} \quad (88)$$

This offers an approach to the hierarchy problem.

.20 Concrete Application: Step by Step

Task: Calculate the Muon Mass

Synergetics method:

1. Determine f_{μ} from tetrahedral geometry ($f_{\mu} = 1/137 \cdot n_{\mu}$)
2. Apply: $m_{\mu} = \frac{1}{f_{\mu}} \times C_{\text{conv}}$
3. Convert to MeV with SI factors
4. Result: 105.1 MeV (0.5% deviation)

T0 method:

1. Logarithmic symmetry: $\ln m_{\mu} = \frac{\ln m_e + \ln m_{\tau}}{2}$
2. Or: $m_{\mu} = \sqrt{m_e \cdot m_{\tau}}$
3. In natural units, with the factor of the lepton ladder: $m_{\mu} = \frac{24\sqrt{3}}{25} \xi^{-1/12} \sqrt{0.511 \times 1777} = 105.4 \text{ MeV (bare)}$ *(Corrected on 30 Sept. 2026: previously $m_{\mu} = \sqrt{0.511 \times 1777} = 105.7 \text{ MeV}$ – the square root alone gives 30.1 MeV; the bare value follows only with the factor of the lepton ladder, -0.26% , Doc. 352.)*
4. Direct, without conversion factors.

In this presentation, the T0 path is shorter.

.21 Philosophical Implications

Both approaches propose a change of perspective:

From	To
Many parameters	One parameter
Empirical	Geometric
Fragmented	Unified
Complicated	Simple
Measurements	Derivations
Big Bang	Static universe [S]

T0 goes one step further:

$$\boxed{\text{Reality} = \text{Geometry} + \text{Time}} \quad (89)$$

In T0, time-mass duality is understood as an **ontological statement** about the nature of reality, not merely as a computational tool.

.22 Numerical Precision: Detailed Comparison

Fundamental Constants

Constant	Synergetics	T0	Experiment	Better
α^{-1}	137.04	137.04	137.036	Equal
$G [10^{-11}]$	6.673	6.6745	6.6743	Equal
$m_e [\text{MeV}]$	0.504	0.511	0.511	T0
$m_\mu [\text{MeV}]$	105.1	105.4 (bare)	105.66	T0
$m_\tau [\text{MeV}]$	1727.6	1777	1776.86	T0
Total	99.0%	99.1%	–	T0

(Corrected on 30 Sept. 2026: T0 value for m_μ previously 105.7 – aligned with the text:
 $\frac{24\sqrt{3}}{25}\xi^{-1/12}\sqrt{m_em_\tau} = 105.4 \text{ MeV (bare, } -0.26\% \text{; Doc. 352).)$

Possible Reasons for the Differences

Why are the T0 values in this table somewhat closer to experiment?

1. **No rounding errors** from unit conversion
2. **Direct geometric relations** without intermediate steps
3. **Logarithmic symmetry** captures subtle structures
4. **Time-mass duality** automatically accounts for relativistic effects

Extension: The Synergetics method derives $1/137$ from α -derived patterns (e.g., $1/\alpha^2 - 1 = 18768$) and builds a bridge to Fuller's geometry. The many floating-point numbers in its calculations and tables (e.g., 7.783×10^{-3} for conversions) make the overview harder, however. In T0, direct formulas like $m_\mu = \frac{24\sqrt{3}}{25}\xi^{-1/12}\sqrt{m_em_\tau}$ lead to shorter calculation paths with fewer error sources. (Corrected on 30 Sept. 2026: previously $m_\mu = \sqrt{m_e \cdot m_\tau}$ – the square root alone gives 30.1 MeV.)

.23 Experimental Distinction

Where Both Theories Make Identical Predictions

- Fine-structure constant
- Gravitational constant
- Most particle masses
- Basic cosmological structure [S]

Where T0 Makes Distinguishable Predictions

Critical tests for T0:

1. **Tau g-2:** $\Delta a_\tau = 7.11 \times 10^{-7}$
 - Synergetics: No prediction
 - T0: Specific value via ξ
2. **Neutrino masses:** $\Sigma m_\nu = 13.6 \text{ meV}$ (*Note of 30 Sept. 2026: excluded by oscillation data – $\Sigma m_\nu \gtrsim 59 \text{ meV}$ for normal ordering, masses not degenerate; see the neutrino section.*)
 - Synergetics: No prediction
 - T0: Specific value
3. **Casimir at $L = 100 \mu\text{m}$:**
 - Synergetics: Not addressed
 - T0: Special resonance
4. **CMB spectrum [S]:**
 - Synergetics: Qualitative
 - T0: Quantitative deviations at high l

.24 Pedagogical Considerations

Synergetics Strengths

- **Visual intuition:** Road map analogy
- **Hands-on:** Buckyballs, physical models
- **Step by step:** From simple to complex
- **Geometric clarity:** IVM structure visible

T0 Strengths

- **Mathematical simplicity:** No additional conversion factors
- **Systematics:** 8 building documents
- **Breadth:** From QM to cosmology
- **Precision:** Concrete numerical predictions

Ideal Teaching Method

Combined approach:

1. **Start:** Synergetics visualizations
 - Understand tetrahedral packing
 - Road map analogy
 - Physical models
2. **Transition:** Introduce natural units
 - Why $c = 1$ makes sense
 - Dimensional analysis
 - Recognize simplification
3. **Deepening:** T0 formalism
 - Time-mass duality
 - Pure geometric derivations with ξ
 - Testable predictions

Extension: This method could be integrated into curricula, starting with Fuller's buckyballs for pupils (visual), followed by T0 formulas for students (analytical).

[S] The statement that pilot studies showed 30% better comprehension rates is not supported in the corpus and would need to be checked.

Assessment of T0 in this document:

- Mathematically simpler
- Physically more far-reaching
- With more concrete numerical predictions
- Conceptually more closed
- More systematically worked out

Extension: T0's strength lies in its predictive power, e.g., the g-2 value, which agrees with the Fermilab data. It offers a bridge to established physics, e.g., through integration into the Standard Model (Yukawa from ξ).

Common Assessment

Both approaches assume:

Nature is geometrically structured

(90)

The fact that two independent approaches arrive at practically identical results is an **indication** that the basic idea could be viable.

T0 adds:

- Time-mass duality as foundation
- Natural units reduce complexity
- Time field as an explanatory approach for anomalies
- Quantitative cosmology without Big Bang **[S]**
- Systematic, testable predictions

Extension: The convergence underscores a “geometric convergence theory”: Independent paths can lead to similar structures, just as Newton and Leibniz arrived at calculus independently. This argues for further testing and invites collaborative extensions, e.g., joint GitHub repositories.

.25 Concluding Remarks

The two independent approaches converge in essential points. The video shows a Synergetics-inspired path with many viable insights. Through the consistent use of natural units and the explicit formulation of time-mass duality, T0 theory gets by with fewer conversion factors and delivers more specific, testable predictions.

The core statement: The geometry of space determines physics, and a single parameter $\xi = \frac{4}{3} \times 10^{-4}$ (corresponding to $1/137$ in Synergetics; plus integer structural choices as motivated settings) is meant to suffice to describe physics; the cosmic sector is left aside (P39).

Extension: Future work could form a “T0-Synergetics alliance,” with joint publications and experiments, e.g., Casimir measurements at ξ lengths. Such measurements would be a direct test of both approaches.

Both approaches arrive at similar structures T0 takes the path via natural units **T0 Theory:**
Time-Mass Duality Framework *Simplicity through natural units*

.26 Bibliography

Bibliography

- [1] Pascher, J. (2025). *T0 Theory: Fundamental Principles*. T0 Document Series, Document 1.
- [2] Pascher, J. (2025). *T0 Theory: The Fine-Structure Constant*. T0 Document Series, Document 2.
- [3] Pascher, J. (2025). *T0 Theory: The Gravitational Constant*. T0 Document Series, Document 3.
- [4] Pascher, J. (2025). *T0 Theory: Particle Masses*. T0 Document Series, Document 4.
- [5] Pascher, J. (2025). *T0 Theory: Neutrinos*. T0 Document Series, Document 5.
- [6] Pascher, J. (2025). *T0 Theory: Cosmology*. T0 Document Series, Document 6.
- [7] Pascher, J. (2025). *T0 Quantum Field Theory: QFT, QM, and Quantum Computers*. T0 Document Series, Document 7.
- [8] Pascher, J. (2025). *T0 Theory: Anomalous Magnetic Moments*. T0 Document Series, Document 8.
- [9] Fuller, R. B. (1975). *Synergetics: Explorations in the Geometry of Thinking*. Macmillan Publishing.
- [10] Winter, D. (2024). *Origins of Gravity and Electromagnetism: Synergetics Insights*. YouTube Transcript (October 28, 2024).
- [11] Feynman, R. P. et al. (1963). *The Feynman Lectures on Physics*. Addison-Wesley.
- [12] Einstein, A. (1917). *Cosmological Considerations on the General Theory of Relativity*. Proceedings of the Prussian Academy of Sciences.
- [13] Planck, M. (1900). *On the Theory of the Energy Distribution Law of the Normal Spectrum*. Proceedings of the German Physical Society.
- [14] Close, F. (1979). *An Introduction to Quarks and Partons*. Academic Press.
- [15] Particle Data Group (2022). *Review of Particle Physics*. Prog. Theor. Exp. Phys. **2022**, 083C01.
- [16] CODATA (2018). *Fundamental Physical Constants*. National Institute of Standards and Technology.
- [17] Weinberg, S. (1995). *The Quantum Theory of Fields, Volume 1*. Cambridge University Press.
- [18] Weinberg, S. (1989). *The Cosmological Constant Problem*. Reviews of Modern Physics, 61(1), 1–23.

- [19] Dirac, P. A. M. (1929). "Quantum mechanics of many-electron systems." *Proc. R. Soc. A* **123**, 714–733. (The quotation originates from this paper; *The Principles of Quantum Mechanics* was published in 1930.)
- [20] KATRIN Collaboration (2022). *Direct Neutrino Mass Measurement with KATRIN*. *Nature Physics*, 18, 474–479.
- [21] LIGO Scientific Collaboration (2016). *Observation of Gravitational Waves*. *Phys. Rev. Lett.* **116**, 061102.
- [22] NumPy Developers (2023). *NumPy Documentation*. Online: <https://numpy.org/doc/>.
- [23] SymPy Developers (2023). *SymPy Documentation*. Online: <https://docs.sympy.org/>.